15 WMW test and Paired t-test

15.1 The Wilcoxon-Mann-Whitney (WMW) Test

The WMW test is used to compare two independent samples like the t-test, but unlike the t-test, the WMW test is valid even when the population distributions are not normal.

more general *

H_0 : The population distributions Y_1 and Y_2 are the same.

H_A : The population distribution of Y_1 is shifted from the population distribution of Y_2.

The WMW test statistic is denoted U_s and measures the degree of separation or shift between the two samples.

The test procedure:

1. Arrange the observations in increasing order
2. Create two counts K_1 and K_2 as follows:
    1. For K_1, count 1 for all observations in sample 2 which are less than each observation in sample 1. Count 0.5 for all observations in sample 2 which are the same value as an observation in sample 1.
    2. For K_2 same thing but in reverse.
    3. $U_s = \max\{K_1, K_2\}$

Example: Soil respiration is a measure of microbial activity in soil, which affects plant growth. In one study, soil cores were taken from two locations in a forest: (1) under an opening in the forest canopy (the "gap" location) and (2) at a nearby area under heavy tree growth (the "growth" location). The amount of carbon dioxide given off by each soil core was measured (in mol CO2/g soil/hr).

H_0 : The gap and growth areas do not differ with respect to soil respiration.

$H_A(nondirectional)$: The distribution of soil respiration rates tends to be higher in one of the two populations.

$H_A(directional)$: Soil respiration rates tend to be greater in the growth area than they are in the gap area.

```
y1<-c(6,13,14,15,16,18,22,29)
y2<-c(17,20,22,64,170,190,315)
```

K_1	Y_1	Y_2	K_2
		17	4
0	6	20	5
0	13.	22	5.5
0	14.	64	8
0	15	170	8
0	16	190	8
1	18	315	8
2.5	22		
3	29		
6.5			49.5

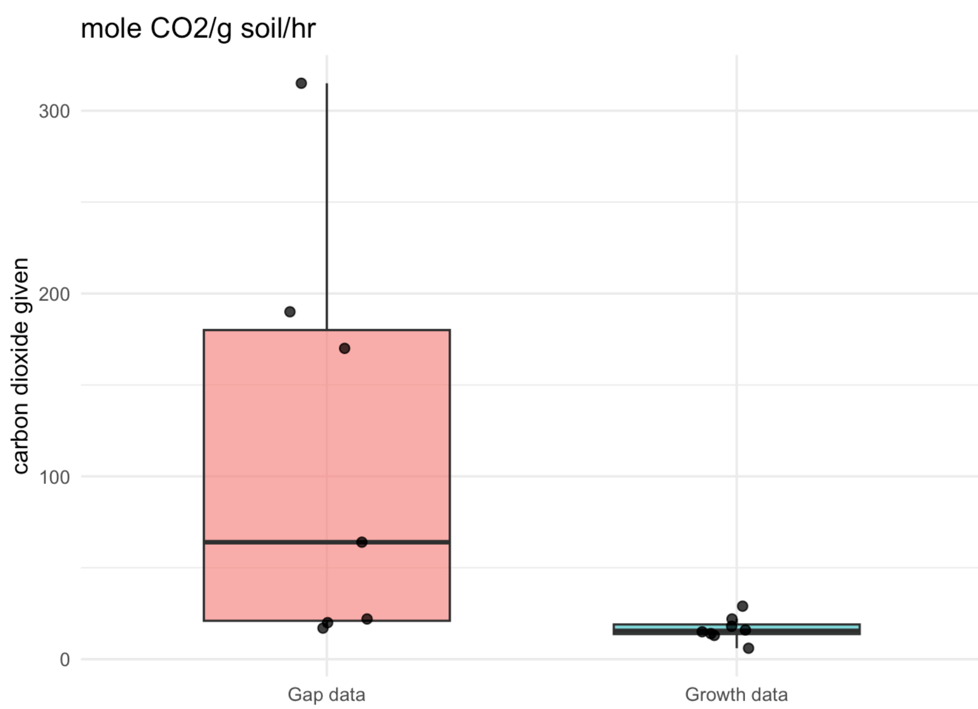
$$U_1 = \max \{K_1 K_2\} = 49.5$$

```
dat<-data.frame(
  group = factor(c(rep("Gap data",length(y2)),rep("Growth data"
  values = c(y2,y1)
)

library(ggplot2)

ggplot(dat, aes(x = group, y = values, fill = group)) +
  geom_boxplot(width = 0.6, outlier.shape = NA, alpha = 0.6) +
  geom_jitter(width = 0.1, height = 0, alpha = 0.8, size = 2) +
  labs(
    x = NULL,
    y = "carbon dioxide given",
    title = "mole CO2/g soil/hr"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "none")
```

← labels for treatment and control




```
w <- wilcox.test(values ~ group, data = dat, exact = FALSE)
w
```

```
    Wilcoxon rank sum test with continuity correction

data:  values by group
W = 49.5, p-value = 0.015
alternative hypothesis: true location shift is not equal to 0
```

```
#one sided
w2 <- wilcox.test(values ~ group, data = dat, alternative="grea
w2
```

```
    Wilcoxon rank sum test with continuity correction

data:  values by group
W = 49.5, p-value = 0.007499
alternative hypothesis: true location shift is greater than 0
```

- The required assumption in the WMW test is that the samples must be random samples from their populations, independently drawn.
- The t-test requires the same assumption of random, independent samples, but it also assumes (1) the normality of the populations or (2) the samples are large enough.
- The MWW test is good for nonsymmetric data.
- The downside is that the statistic does not use all information in the data, only ranks and so it loses statistical power.

15.2 The paired-sample t-test and CI

So far we have assumed that the we have two independent samples, one from each population. We will now consider the comparison of two samples that are not independent bu are **paired**. In a paired design, the observations (Y_1, Y_2) occur in pairs; the observational units in a pair are linked.

The 2 samples are not independent.

Pairing in an experimental design can serve to reduce bias, to increase precision, or both. Usually the primary purpose of pairing is to increase precision.

For example: consider testing whether a fertilizer increases plant growth using the same plants measured twice.

12 tomato plants are measured for height after 2 weeks under standard conditions. Then all ~~12~~ 6 plants receive a new fertilizer for 2 more weeks and height is measured again. Because each plant is measured before and after, observations are paired.

$$H_0 : \mu_D = \mu_1 - \mu_2 = 0$$

$$H_A : \mu_D = \mu_1 - \mu_2 \neq 0.$$

$\underbrace{SE(\bar{x}-\bar{y})}_{\text{paired experiment}} \leq \underbrace{SE(\bar{x}-\bar{y})}_{\text{independent samples}}$

```r
# Paired t-test example data (plant heights in cm)
dat <- data.frame(
  plant  = 1:6,
  before = c(12.1, 11.8, 13.0, 10.9, 12.5, 11.2),
  after  = c(14.0, 13.2, 14.1, 12.0, 13.7, 12.6)
)

dat
```

	plant	before	after	d
1	1	12.1	14.0	1.9
2	2	11.8	13.2	2.4
3	3	13.0	14.1	.
4	4	10.9	12.0	.
5	5	12.5	13.7	:
6	6	11.2	12.6	
		$\bar{X}$	$\bar{Y}$	$\bar{D}$

← one-sa

$SE_D \neq SE_{\bar{X}-\bar{Y}}$

$\bar{D} = \bar{Y} - \bar{X}$

$SE_{\bar{D}} = \cancel{\sqrt{SE_X^2 + SE_Y^2}}$

$= \sqrt{\left(\frac{SD_X}{\sqrt{n}}\right)^2 + \left(\frac{SD_Y}{\sqrt{n}}\right)^2}$

The test statistic is

$$\frac{\bar{D}-0}{SE_{\bar{D}}} = \frac{\bar{D}-0}{SD_D/\sqrt{n}}$$

which is distributed as a t-distribution with $df = n-1$.

A $(1-\alpha)$ confidence interval for $\mu_D is \bar{D} \pm t_{n-1,\alpha/2} SE_{\bar{D}}$\$ In our example this is:

```
# CI by hand
d<-mean(dat$after-dat$before) ←
sd<-sd(dat$after-dat$before)
se<-sd/sqrt(6)
se
```

```
[1] 0.123153
```

```
tval<-qt(.975,5)
d-tval*se
```

```
[1] 1.033425
```

```
d+tval*se
```

```
[1] 1.666575
```

```
print(paste("95% CI is: (",d-tval*se,d+tval*se,")"))
```

```
[1] "95% CI is: ( 1.03342508032405 1.66657491967595 )"
```

```
# Paired t-test
t.test(dat$after, dat$before, paired = TRUE)
```

```

    Paired t-test

data:  dat$after and dat$before
t = 10.962, df = 5, p-value = 0.0001099
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
```

```
 1.033425 1.666575
sample estimates:
mean difference
           1.35
```

What is the difference between a paired and an unpaired test?

```
# CI by hand
d<-mean(dat$after-dat$before)
se1<-sd(dat$after)/sqrt(6)
se2<-sd(dat$before)/sqrt(6)
se_unpooled<-sqrt(se1^2+se2^2)
df<-((se1^2+se2^2)^2)/(((se1^2)^2 / 5)+((se2^2)^2 / 5))
tval<-qt(.975,df)
d-tval*se
```

```
[1] 1.075484
```

```
d+tval*se
```

```
[1] 1.624516
```

```
print(paste("95% CI is: (",d-tval*se_unpooled,d+tval*se_unpooled
```

```
[1] "95% CI is: ( 0.305927531597175 2.39407246840282 )"
```

```
# unpaired two-sided
t.test(dat$after, dat$before, paired = FALSE)
```

```
    Welch Two Sample t-test

data:  dat$after and dat$before
t = 2.8822, df = 9.9693, p-value = 0.01637
alternative hypothesis: true difference in means is not equal
to 0
95 percent confidence interval:
 0.3059275 2.3940725
sample estimates:
mean of x mean of y
 13.26667  11.91667
```

A paired experiment reduces the noise, i.e., variation in outcomes due to other variables.